Example Applications

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Outline

Example 1

K-Means Clustering using MapReduce

Example 2

Building a Video-Sharing SaaS Cloud Application

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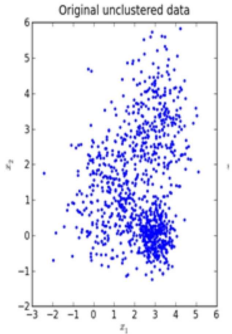
K-Means Clustering

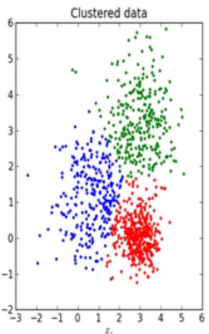
Algorithm to cluster **m objects** based on **n attributes** into **k groups**, where $\mathbf{k} < \mathbf{m}$

Minimise distances between intra-cluster points

Maximise distances between inter-cluster points

Useful method in data analysis (e.g., retail, streaming services, sports science, etc.)





Distance Metric

Three properties

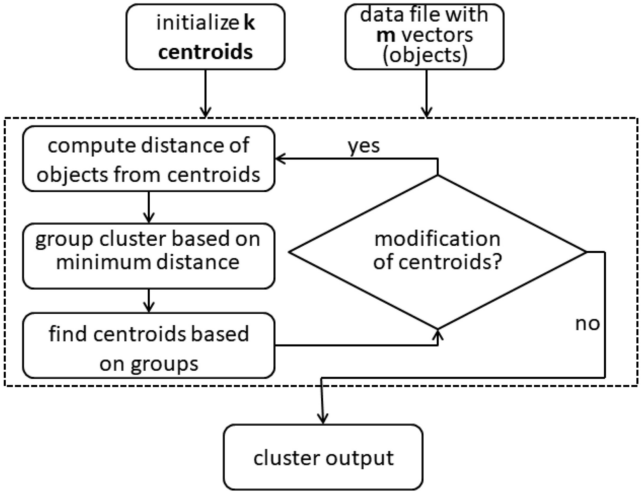
1. Non-negative
2. Equal to 0 iff $p = q$
3. Triangle inequality, i.e., for 3 points p, q and r, d (p, r) $<$ d (p, q) $+$ d (q, r)

Choice of different distance metrics

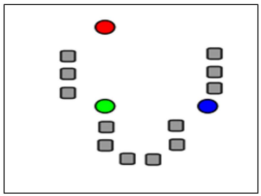
One class is the p-norm

- For any p > 0, distance between vectors q and r (each of dimension n) is defined as $((q_1 - r_1)^p + (q_2 - r_2)^p + \ldots + (q_n - r_n)^p)^{\frac{1}{p}}$
- E.g., Euclidean distance $(2 - \textit{norm distance})$

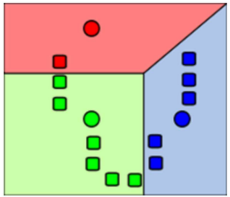
K-Means Algorithm I



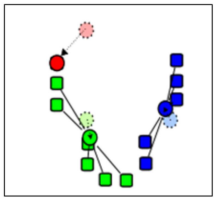
K-Means Algorithm II



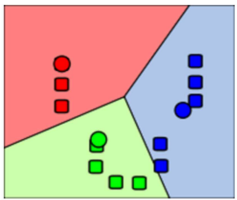
1. initialization



2. compute distance and form clusters



3. recompute centroids



4. output

K-Means Algorithm III

Steps

1. Initialise
2. Iterate
   - Compute distance and form clusters
   - Recompute centroids
3. Output results

Challenge!

What is **k**?

Dataset

Twitter dataset from `https://snap.stanford.edu/data/egonets-Twitter.html`

Aim

Apply k-means to determine how many Twitter users fall under a cluster with a centroid of X followers and Y followees

Using MapReduce

What is the **Map** phase? – compute clusters

What is the **Reduce** phase? – compute new centroids

Map Phase

Compute distances between an input point and all the previously obtained centroid of clusters

Find the cluster with the minimum distance

Add the input point into the cluster with the minimum distance

Reduce Phase

Receive all the points belonging to a particular cluster

Compute new centroid

Input Key-Value Pairs I

Important!

Pre-process the data!

Key $\rightarrow$ ID of a user (or participating point)

Value $\rightarrow$ vector of attributes of that particular ID

Example

key: 104997734; value: len: 2; 0:1.0; 1:2.0;

key: 46198765; value: len: 2; 0:5.129283016944966; 1:2.321928094887362;

"len: 2" $\Rightarrow$ two elements

num followers num followees

"0:5.12" $\Rightarrow 0^{th}$ element is 5.12 $\Leftrightarrow \log_2$ (number of followers)

"1:2.32" $\Rightarrow 1^{st}$ element is 2.32 $\Leftrightarrow \log_2$ (number of followees)

Input Key-Value Pairs II

Why log?

Deal with outliers in the data

E.g., if someone has 20,000 followers, the clusters are heavily affected and hence, common practice is to take log values to dampen this type of effect

Output Key-Value Pairs

Key $\rightarrow$ ID of a cluster

Value $\rightarrow$ vector representing centroid for the cluster and the number of points inside the cluster

Example

key: cluster0; value: 23904; len 2; 0:2.005394687254275; 1:1.3290466203187175;

key: cluster1; value: 22125; len 2; 0:5.281555290448195; 1:5.6107313297218275;

x coords y coords

K-Means Implementation I

Steps

1. Upload data to cloud
2. Upload executable code to cloud
3. Create clusters
4. View output results on cloud
5. Repeat as necessary, to get next clusters

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K-Means Implementation II

Technology Needed!

Storage (e.g., S3)

- Amazon S3 (Simple Storage Service) is a web **storage** service

MapReduce Engine (e.g., Amazon EMR)

- Amazon EMR (previously called Amazon Elastic MapReduce) is a managed cluster platform that simplifies running big data frameworks

K-Means Implementation III

AWS Example



Data and code loaded on S3 storage

Implementation using EMR

Output results stored on S3

K-Means Example Output I

key: cluster0; value: 23904; len 2; 0:2.005394687254275; 1:1.3290466203187175;

key: cluster1; value: 22125; len 2; 0:5.281555290448195; 1:5.6107313297218275;

key: cluster2; value: 5166; len 2; 0:5.051385274987691; 1:2.7115082846455305;

key: cluster3; value: 25074; len 2; 0:2.944391259561573; 1:3.7132202620068284;

What can we deduce?

K-Means Example Output II

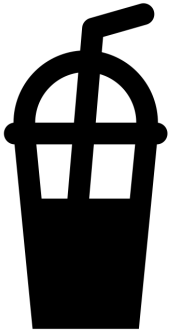
Number of points in the cluster that has a centroid with $\log_2$ (number of followers) = 5.05 and, $\log_2$ (number of followees) = 2.71 is only **5166**

Other clusters with small differences between number of followers versus number of followees contain many more points

Verdict

A user typically has a similar number of followers and followees

Take a Break







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Objective

Aim

Demonstrate the building of a SaaS video-sharing application

Users can **upload** (store) videos

Users can **stream** or **download** videos

Challenges

Videos are uploaded in different formats ⇒ convert them to streaming format

Video conversion – CPU intensive!

not every format can be streamed..

Upload of videos and streaming videos to multiple users – I/O intensive!

What about user accounts? For now, no (thankfully! ☺)

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Video-Sharing Application

Steps

1. Upload shared videos (**Upload Process**)
2. Encode uploaded videos (**Encoding Process**)
3. Stream shared videos to users (**Streaming Process**)

Upload Process I

Requirements

Cloud consumer uses a web browser

SaaS Provider

- Interface to upload videos
- Storage to store videos
- Need to balance workload

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Upload Process II

Technology Needed!

Web servers – to manage client connections (e.g., EC2)

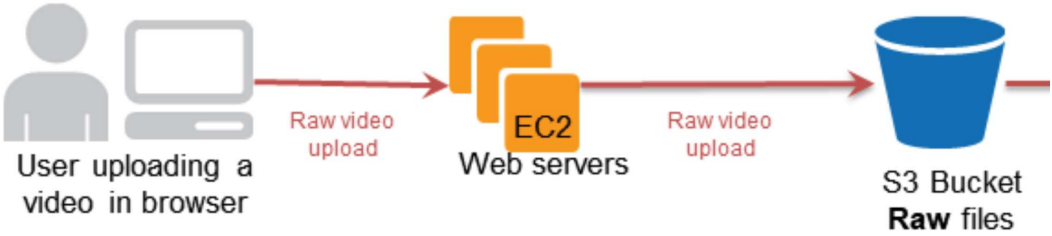
- Amazon EC2 (Elastic Compute Cloud) is a web service that enables launching and managing *server instances* in Amazon's data centers

Storage servers (e.g., S3)

- Amazon S3 (Simple Storage Service) is a web **storage** service

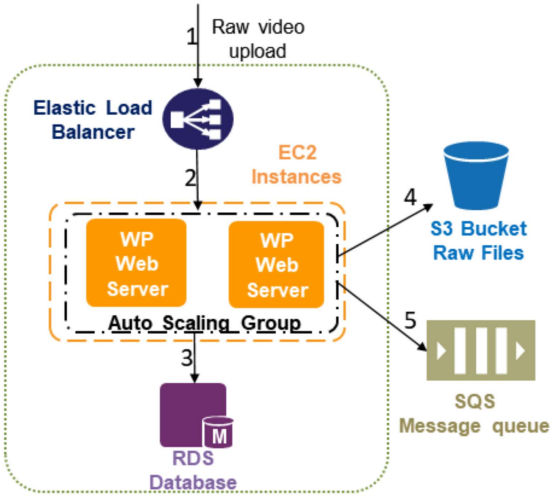
Additional elements could include a load-balancer, auto-scaling, message passing, database, etc.

Upload Process III



Upload Process

Detailed



machine level

need to program some logic

Name	IaaS	PaaS	SaaS
EC2	Y		
S3	Y		
Auto scaling	Y		
ElasticLoadBalancer	Y		
SQS		Y	
Web Browser			Y

Encoding Process I

Requirements

SaaS Provider

- Video encoder
- Storage to store encoded videos
- Need to balance workload (because encoding is compute expensive)

Encoding Process II

Technology Needed!

Video encoder (e.g., ffmpeg)

- ffmpeg is an open source project, which allows for video transcoding

Servers – to manage the encoding (e.g., EC2)

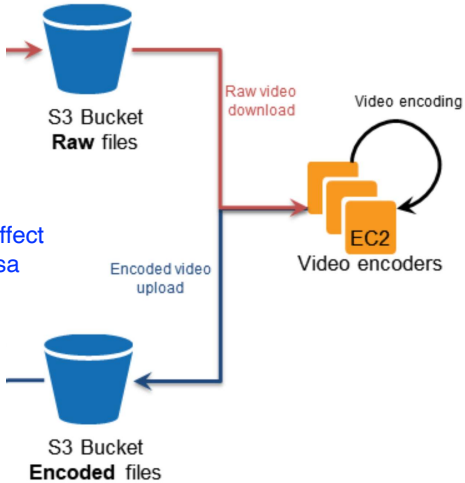
Storage servers (e.g., S3)

Additional elements could include auto-scaling, message passing, etc.

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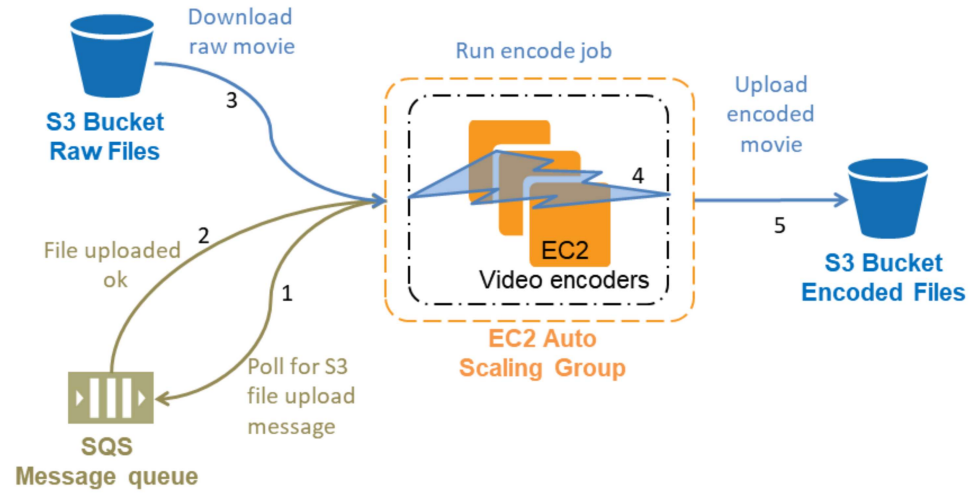
Encoding Process III

why separate?
so busy uploading wouldnt affect the encoding and vice versa



Encoding Process

Detailed



Name	IaaS	PaaS	SaaS
EC2	Y		
S3	Y		
Auto scaling	Y		
ElasticLoadBalancer	Y		
SQS		Y	
Video Encoder		Y	

internal component for dev

Requirements

Cloud consumer uses a web browser

SaaS Provider

- Video streamer
- Storage for encoded videos (shared with previous stage)
- Need to balance workload

Streaming Process II

Technology Needed!

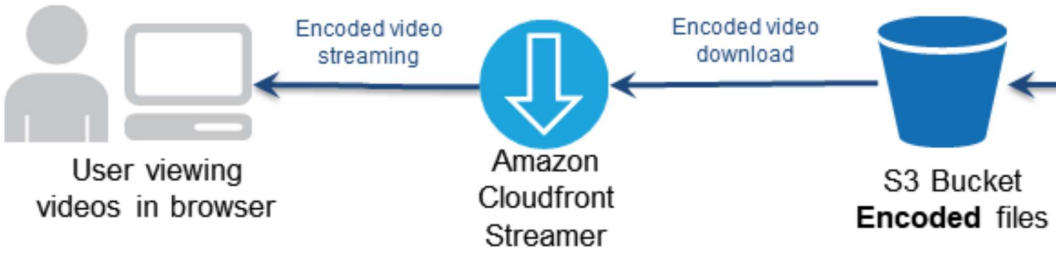
Streaming servers – to stream the videos (e.g., Amazon CloudFront Streamer)

- Amazon CloudFront is a web service that speeds up (content) distribution of static and dynamic web content through a worldwide network of edge locations

local cache

Storage servers (e.g., S3)

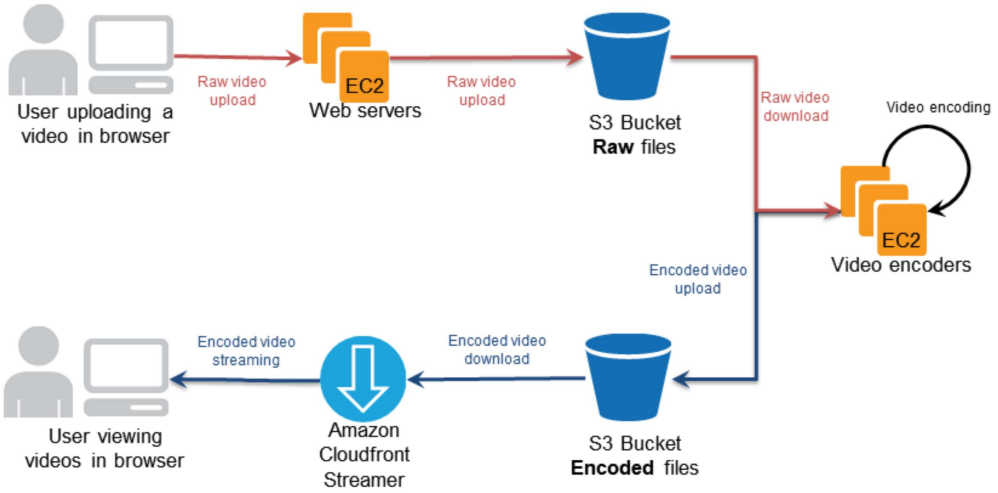
Streaming Process III



for the end user

Name	IaaS	PaaS	SaaS
S3	Y		
CloudFront			Y
Media Player			Y
Entire Application			Y

Putting It All Together



Pricing

Costs

EC2 Instances

S3 storage

CloudFront content delivery network

PaaS software – based on usage

Pricing Example

≈ 50 uploading hits per day

- Each upload is 100 MB of HD movie
- Encoding process takes 1 minute per movie
- 5 GB per day or 150 GB per month

≈ 500 streaming hits per day

- Each user streams 100 MB of HD movies (≈ 3 minutes)
- Assume same location
- 50 GB per day or 1.5 TB per month

Can use AWS Pricing Calculator – `https://calculator.aws`

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Pricing Comparisons

Traditional

Compare cloud vs on-premise

Inter-Cloud

Compare between different cloud providers

Intra-Cloud

Compare between different offerings with the same provider (e.g., different EC2 instances, etc.)

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Next Week

Infrastructure as Code

Continuous Feedback



https://forms.cloud.microsoft/r/ZJxTf2wFjr

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